Magic and Wormholes

Brian Swingle (Brandeis)
IAS Seminar, March 23, 2026

With Valerie Bettaque (2602.12339)
and Alejandro Vilar Lopez and
Martin Sasieta (2512.00149)

Alchemical ouroboros (Wikipedia)

Valerie Bettaque

Javier Magan

Martin Sasieta

physicsmonkey.github.io

We consider a system of N Majorana fermions χ_i

Majorana strings, a complete basis of Hermitian operators: $\mu(a) \propto \chi_1^{a_1} \cdots \chi_N^{a_N}$

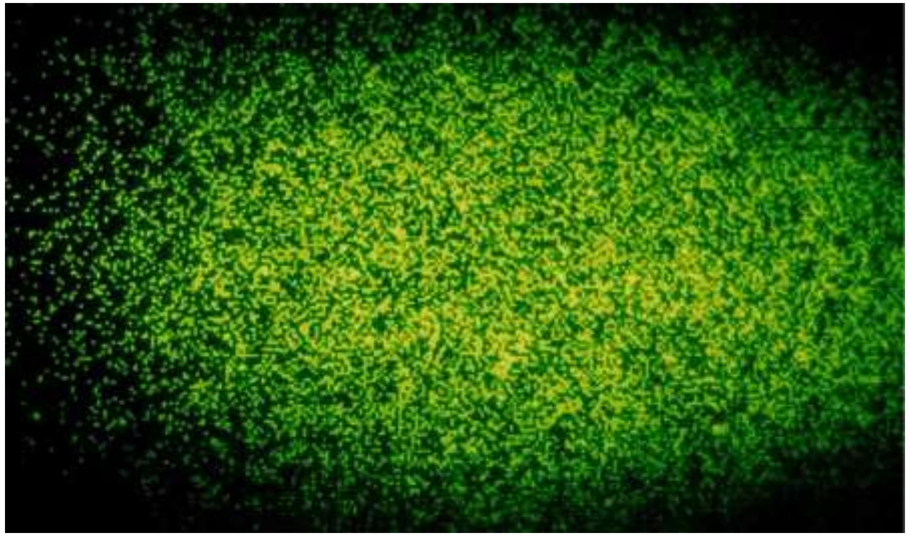
Thermal 1-point functions, these completely determine the state: $\xi(a) = \frac{\mathrm{tr}\left(e^{-\beta H}\mu(a)\right)}{\mathrm{tr}\left(e^{-\beta H}\right)}$

In this talk, we will consider ensembles of Hamiltonians and investigate the statistical properties of these 1-point functions

Physics of 1-point functions

1. They completely determine the quantum state and hence go far beyond standard few-body measurements, but there is a lot of data to make sense of (exponential in N)
2. This data is useful for understanding:
   A. Information scrambling
   B. Snapshots, classical shadows

$$e^{iHt}\mathrm{V}e^{-iHt} = \sum_{a} \mathrm{v}(\mu, t)\mu(a)$$



[Greiner lab]

Physics of 1-point functions

1. They completely determine the quantum state and hence go far beyond standard few-body measurements, but there is a lot of data to make sense of (exponential in N)
2. This data is useful for understanding:
   A. Information scrambling
   B. Snapshots, classical shadows
3. Today we'll discuss how it relates to:
   A. Quantum magic, primarily in the context of SYK
   B. Wormholes and closed universes

Sachdev-Ye-Kitaev model (q=4)

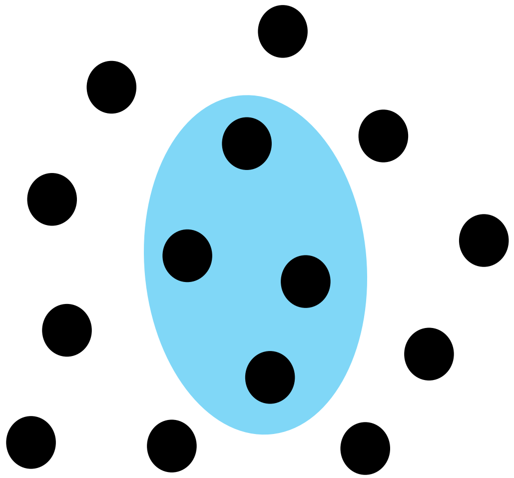
$$\chi_a^\dagger = \chi_a,\ a = 1, \cdots, N$$

$$\{\chi_a, \chi_b\} = \delta_{a,b}$$

$$D = 2^{N/2}$$

$$H = \frac{1}{4!}\sum_{abcd} J_{abcd}\chi_a\chi_b\chi_c\chi_d$$

$$\overline{J_{abcd}} = 0,\ \overline{J_{abcd}^2} = \frac{3!J^2}{N^3}$$



[random 2-body ensemble, Sachdev-Ye, Kitaev, Rosenhaus-Polchinski, Maldacena-Stanford, Garcia-Garcia-Verbaarschot, Bagrets-Altland-Kamenev, ...]

Plan

1. Review the notion of magic as a resource and recent developments in many-body magic
2. The computation of magic measures in the SYK model
   A. Thermal 1-point functions behave as real Gaussian random variables for $q>2$ [Bettaque-S]
3. The relation between these SYK computations and wormhole computations in holography
   A. SYK calculation can be fit with a gravity calculation involving a 2d wormhole stabilized by heavy matter with N light fields [Bettaque-S]
4. Connections to recent models of closed universes [Sasieta-S-Vilar Lopez]

Quantum magic

Resources in quantum theory

- One useful way to view physical theories is in terms of resources, e.g. quantum entanglement
  A. "Free states" = unentangled states
  B. Resource measures = entanglement entropy, …
- Entanglement is necessary for teleportation, multi-player quantum games, and general quantum computations, and it is related to the geometry of spacetime in holography
- From the perspective of classical simulation, low entanglement offers an opportunity that we can take advantage of with e.g. tensor networks
- Magic is the resource theory of "non-stabilizerness"

Fermionic Cliffords and Stabilizers

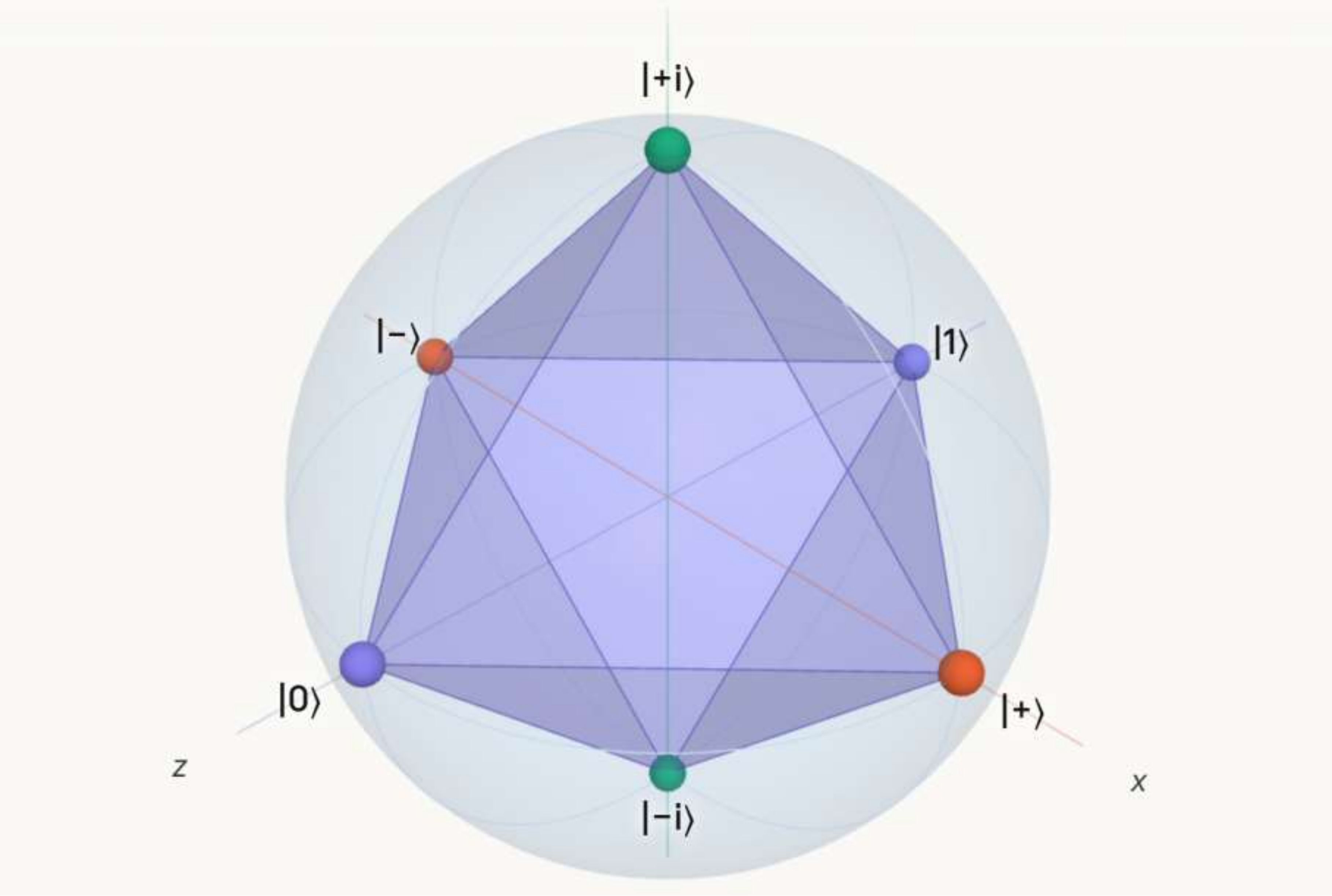
Clifford group maps strings to strings, no superposition: $\mu(a) \to U\mu(a)U^\dagger \propto \mu(a')$

Stabilizer states are obtained by acting on a fixed product basis $\left(2\mathrm{i}\chi_{2\mathrm{j}-1}\chi_{2\mathrm{j}} = 1\right)$ with Cliffords, which imparts classical simulability

Recently, we proved a number of facts about this group, which turns out to be $O(N, Z_2)$ after demanding fermion parity symmetry; we also showed that it forms 3-design and gave an efficient sampling algorithm

[Bettaque-S 2407.11319]

Single qubit stabilizer polytope (STAB = mixtures of stabilizer states):



Measures of magic

- There are many measures of magic, which tend to involve analogues of distance to the stabilizer polytope (STAB)
- Many of these are not efficiently computable, and the situation is also harder when dealing with mixed states
- We will focus for simplicity on one measure, robustness of magic

$$R(\rho) = \min\left(\sum_{\alpha} |c_\alpha| : \rho = \sum_{\alpha} c_\alpha \sigma_\alpha \,, \sigma_\alpha \in \mathrm{STAB}\right)$$

$$R(\rho) \geq \frac{F_1(\rho)}{D}, F_1(\rho) = \sum_{a} |\xi(a)|$$

Many-body magic

- The idea of magic is relatively old, going back to Bravyi-Kitaev, and is well-studied in the quantum information community
- In 2020, many-body magic was studied for the first time in the 3-state Potts model [Cao-S-White] and beyond [Liu-Winter]
- There is now an explosion of interest, including the development of novel measures such as stabilizer Renyi entropy [Leone-Oliviero-Hamma, …], magic in SYK [Bera-Schiro, Jasser-Odavic-Hamma, Zhang-Zhou-Sun, …], and proposed connections to gravitational backreaction in holography [Cao-Cheng-Hamma-Leone-Munizzi-Oliviero]

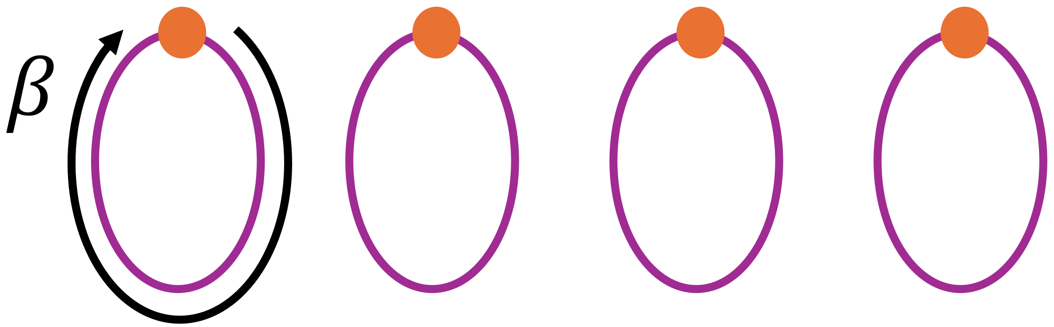
Magic in SYK

Statistical moments

- Many magic measures are built from the Majorana (or Pauli) string data, e.g. robustness and stabilizer renyi entropy

$$F_R = \sum_a |\xi(a)|^R$$

- We will compute the ensemble average of the R-norm with a path integral computation on R replicas



$$R = 4$$

Path integral computation

$$\begin{aligned}\mathrm{tr}\left(\mu(a)\, e^{-\beta H_{\mathrm{SYK}}}\right)^R &= \mathrm{tr}\left[\mu^{\otimes_f R} \exp(-\beta H_{\mathrm{SYK}})^{\otimes_f R}\right] \\ &= \mathrm{tr}\left[\mu^{(1)}(a)\, \mu^{(2)}(a) \cdots \mu^{(R)}(a) \exp\left(-\beta \sum_{r=1}^{R} H_{\mathrm{SYK}}^{(r)}\right)\right] \\ &\simeq \mathrm{tr}\left[(-1)^{a_1 F_{R,1}} \cdots (-1)^{a_N F_{R,N}} \exp\left(-\beta \sum_{r=1}^{R} H_{\mathrm{SYK}}^{(r)}\right)\right]\end{aligned}$$

$$(-1)^{F_{R,i}} := (2i)^{R/2}\, \chi_i^{(1)} \cdots \chi_i^{(R)}$$

[Bettaque-S]

Path integral computation

multi-replica collective field

$$G_{rs}(\tau,\tau') := \frac{1}{N}\sum_{i=1}^{N} \chi_i^{(r)}(\tau)\,\chi_i^{(s)}(\tau'), \quad G_{rs}(\tau,\tau') = -G_{sr}(\tau',\tau),$$

$$\overline{\mathrm{tr}(\mu(a)\, e^{-\beta H_{\mathrm{SYK}}})^R}$$

W = weight = # of non-identity ops in μ

$$= \int [\mathcal{D}\Sigma]\,[\mathcal{D}G]\,\mathrm{pf}[I_R \otimes \mathcal{D}^- - \Sigma]^{N-W}\,\mathrm{pf}[I_R \otimes \mathcal{D}^+ - \Sigma]^W$$
$$\exp\left(-\frac{N}{2}\sum_{r,s=1}^{R}\int_0^\beta d\tau d\tau' \left(\Sigma_{rs}(\tau,\tau')\,G_{rs}(\tau,\tau') - \frac{J^2}{q}G_{rs}(\tau,\tau')^q\right)\right)$$

ensemble average washes out the dependence
on which fermions were chosen for μ

[Bettaque-S]

Path integral computation

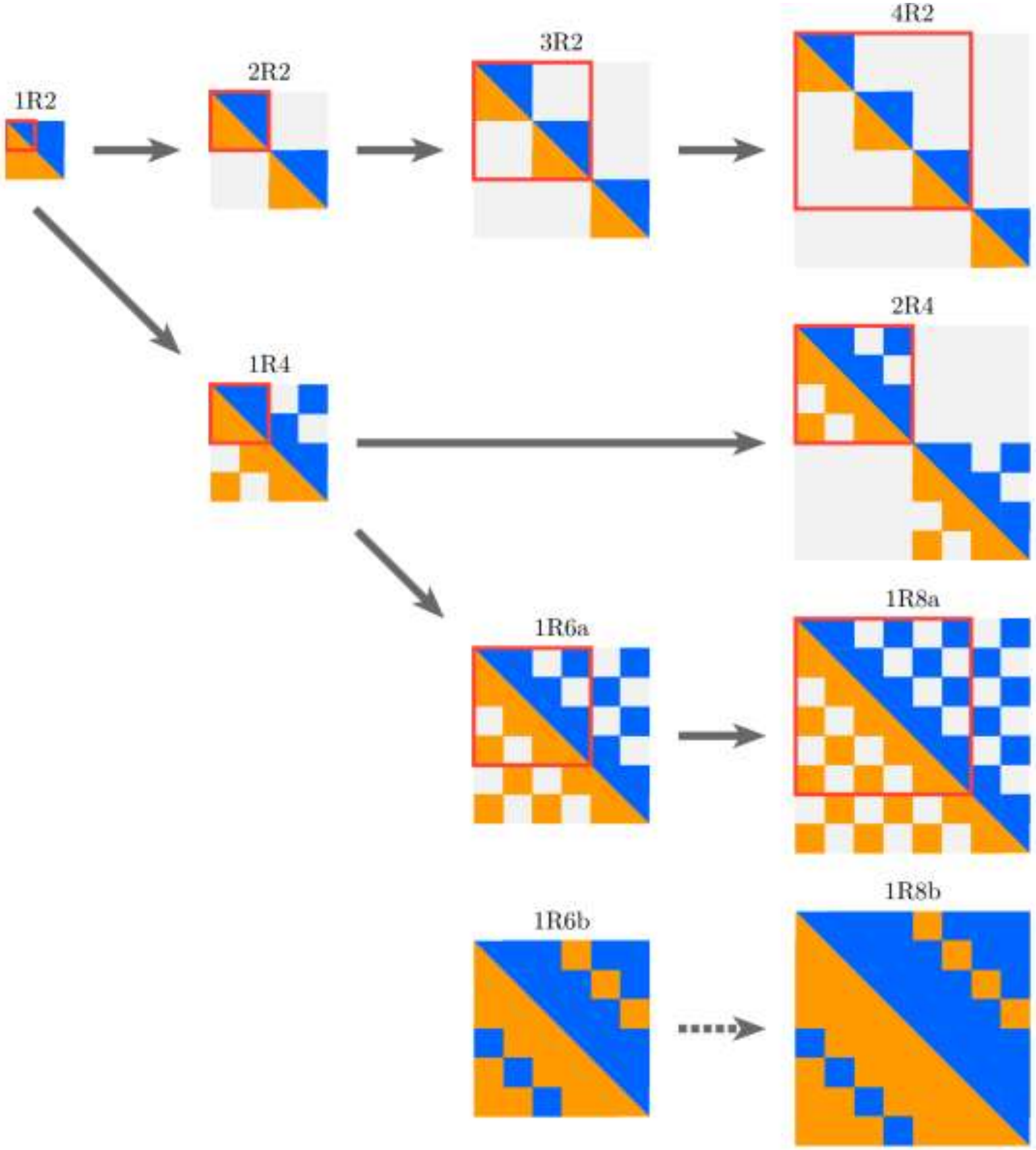
- We evaluate the path integral at large N by finding the leading saddles, but the numerics are demanding (replicas, broken time-translation, ...)

$$\mathcal{I}^{(R,w)}_{\mathrm{SYK}}[G,\Sigma] = -(1-w)\ln \mathrm{pf}[I_R\otimes \mathcal{D}^- - \Sigma] - w\ln \mathrm{pf}[I_R\otimes \mathcal{D}^+ - \Sigma] + \frac{1}{2}\sum_{r,s=1}^{R}\int_0^\beta d\tau d\tau' \left(\Sigma_{rs}(\tau,\tau')\,G_{rs}(\tau,\tau') - \frac{J^2}{q}G_{rs}(\tau,\tau')^q\right)$$

$$\overline{\xi(w)^R} \approx \exp\left(-N\left[\mathcal{I}^{(R,w)}_{\mathrm{SYK}} - R\,\mathcal{I}^{(1,0)}_{\mathrm{SYK}}\right]\right)$$

$$\mathcal{D}^{\mp} = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & \pm 1 \\ -1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & -1 & 1 \end{pmatrix}$$

R	Structure	G_1	G_2	G_3	G_4	G_5	G_6	G_7	Comments
2	$\mathbf{R}_2(+)$	$\mathbb{1}/2$							
4	$\mathbf{R}_4(0,+)$	0	$\mathbb{1}/2$	0					$\cong 2\times\mathbf{R}_2(+)$
	$\mathbf{R}_4(+,0)$	$\mathbb{1}/2$	0	$\mathbb{1}/2$					
	$\mathbf{R}_4(+,+)$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$					$\rightsquigarrow \mathbf{R}_4(+,0)$ limit. convergence
6	$\mathbf{R}_6(0,0,+)$	0	0	$\mathbb{1}/2$	0	0			$\cong 3\times\mathbf{R}_2(+)$
	$\mathbf{R}_6(0,+,0)$	0	$\mathbb{1}/2$	0	$\mathbb{1}/2$	0			limit. convergence
	$\mathbf{R}_6(0,+,+)$	0	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	0			$\rightsquigarrow \mathbf{R}_6(+,+,-)$
	$\mathbf{R}_6(+,0,0)$	$\mathbb{1}/2$	0	0	0	$\mathbb{1}/2$			$\rightsquigarrow \mathbf{R}_6(+,0,+)$
	$\mathbf{R}_6(+,0,+)$	$\mathbb{1}/2$	0	$\mathbb{1}/2$	0	$\mathbb{1}/2$			
	$\mathbf{R}_6(+,0,-)$	$\mathbb{1}/2$	0	$-\mathbb{1}/2$	0	$\mathbb{1}/2$			limit. convergence
	$\mathbf{R}_6(+,+,0)$	$\mathbb{1}/2$	$\mathbb{1}/2$	0	$\mathbb{1}/2$	$\mathbb{1}/2$			$\rightsquigarrow \mathbf{R}_6(+,+,-)$
	$\mathbf{R}_6(+,+,+)$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$			$\rightsquigarrow \mathbf{R}_6(+,0,+)$
	$\mathbf{R}_6(+,+,-)$	$\mathbb{1}/2$	$\mathbb{1}/2$	$-\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$			
8	$\mathbf{R}_8(0,0,0,+)$	0	0	0	$\mathbb{1}/2$	0	0	0	$\cong 4\times\mathbf{R}_2(+)$
	$\mathbf{R}_8(0,0,+,0)$	0	0	$\mathbb{1}/2$	0	$\mathbb{1}/2$	0	0	$\rightsquigarrow \mathbf{R}_8(+,0,+,0)$
	$\mathbf{R}_8(0,0,+,+)$	0	0	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	0	0	$\rightsquigarrow \mathbf{R}_8(+,-,+,+)$
	$\mathbf{R}_8(0,+,0,0)$	0	$\mathbb{1}/2$	0	0	0	$\mathbb{1}/2$	0	$\cong 2\times\mathbf{R}_4(+,0)$
	$\mathbf{R}_8(0,+,0,+)$	0	$\mathbb{1}/2$	0	$\mathbb{1}/2$	0	$\mathbb{1}/2$	0	$\rightsquigarrow \mathbf{R}_8(0,+,0,0)$
	$\mathbf{R}_8(0,+,+,0)$	0	$\mathbb{1}/2$	$\mathbb{1}/2$	0	$\mathbb{1}/2$	$\mathbb{1}/2$	0	$\rightsquigarrow \mathbf{R}_8(+,+,+,-)$
	$\mathbf{R}_8(0,+,+,+)$	0	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	0	$\rightsquigarrow \mathbf{R}_8(+,+,+,-)$
	$\mathbf{R}_8(0,+,+,-)$	0	$\mathbb{1}/2$	$\mathbb{1}/2$	$-\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	0	$\rightsquigarrow \mathbf{R}_8(+,+,+,-)$
	$\mathbf{R}_8(+,0,+,0)$	$\mathbb{1}/2$	0	$\mathbb{1}/2$	0	$\mathbb{1}/2$	0	$\mathbb{1}/2$	
	$\mathbf{R}_8(+,0,+,+)$	$\mathbb{1}/2$	0	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	0	$\mathbb{1}/2$	$\rightsquigarrow \mathbf{R}_8(+,0,+,0)$
	$\mathbf{R}_8(+,+,+,0)$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	0	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\rightsquigarrow \mathbf{R}_8(+,+,+,-)$
	$\mathbf{R}_8(+,+,+,+)$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\rightsquigarrow \mathbf{R}_8(+,0,+,0)$
	$\mathbf{R}_8(+,+,+,-)$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	$-\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	$\mathbb{1}/2$	

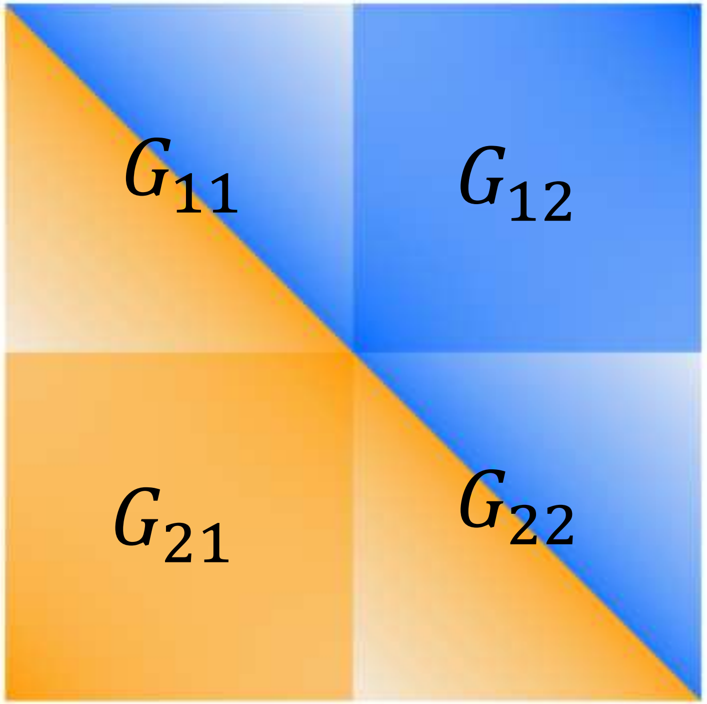


Path integral results

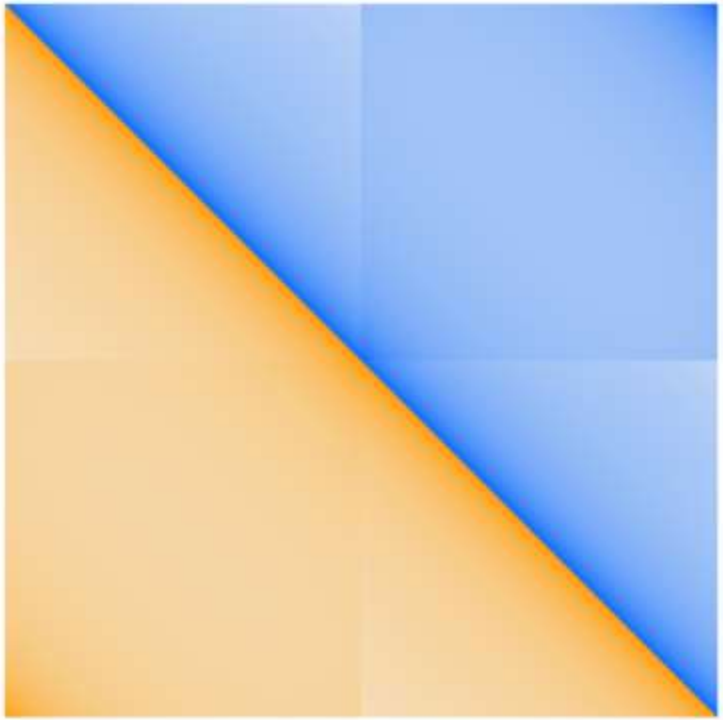
R=4

- We find the 1-point functions are real gaussian at large N provided q>2 (SYK interaction degree)

R=2

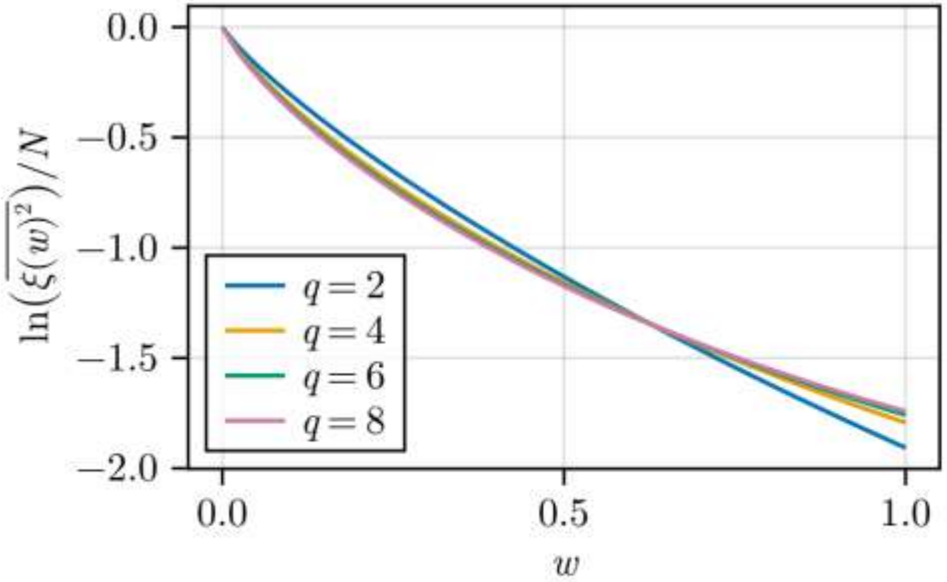


(b) $\beta = 5, w = 0.5$

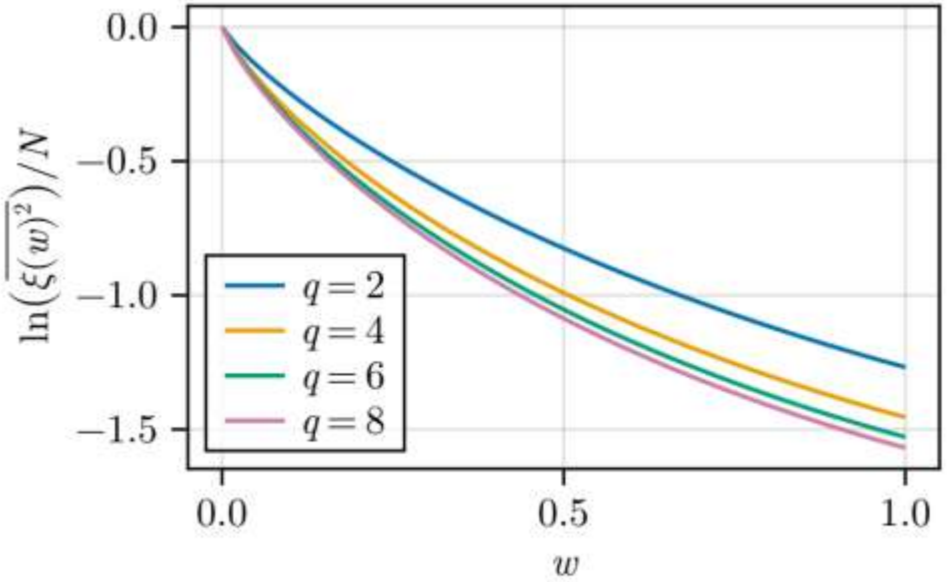


(e) $\beta = 20, w = 0.5$

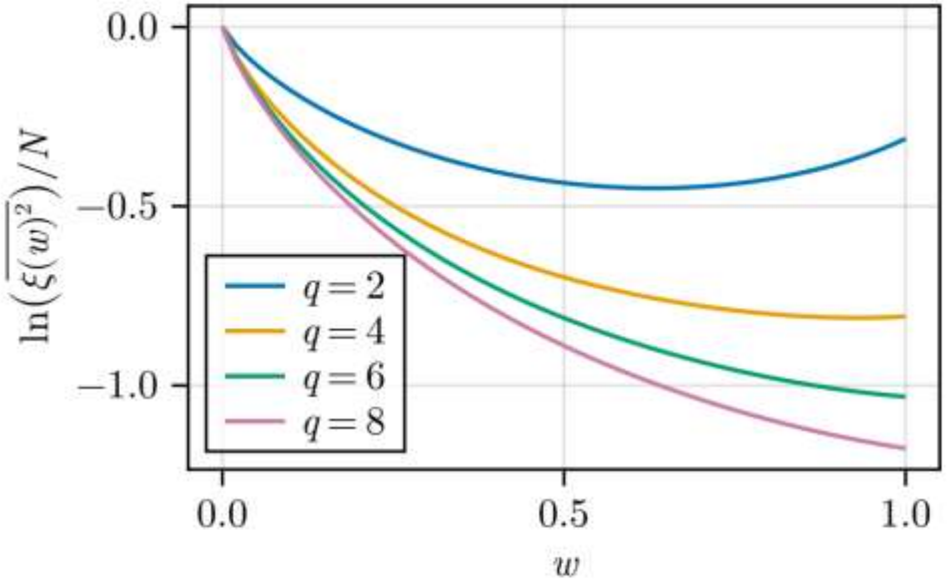
[Bettaque-S]



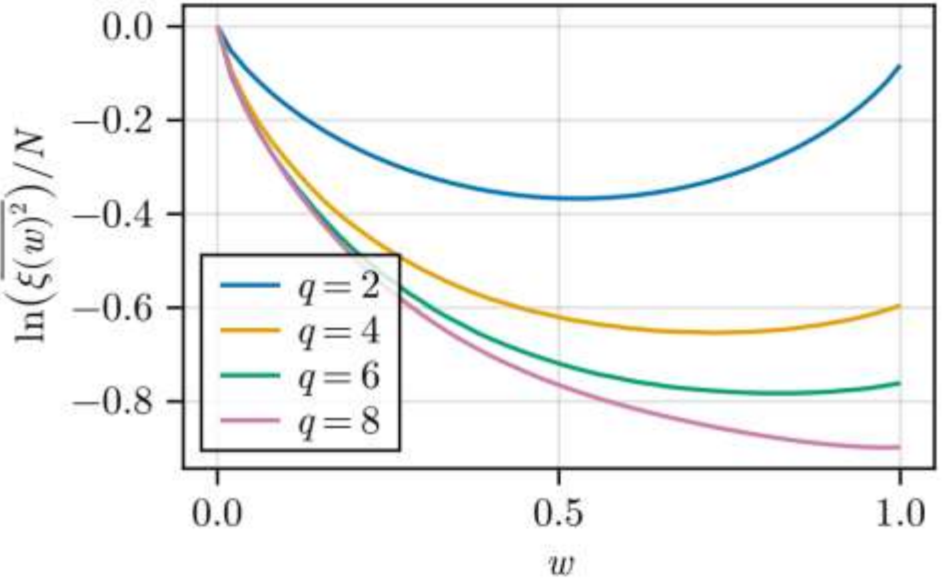
(a) $\beta = 0.5$



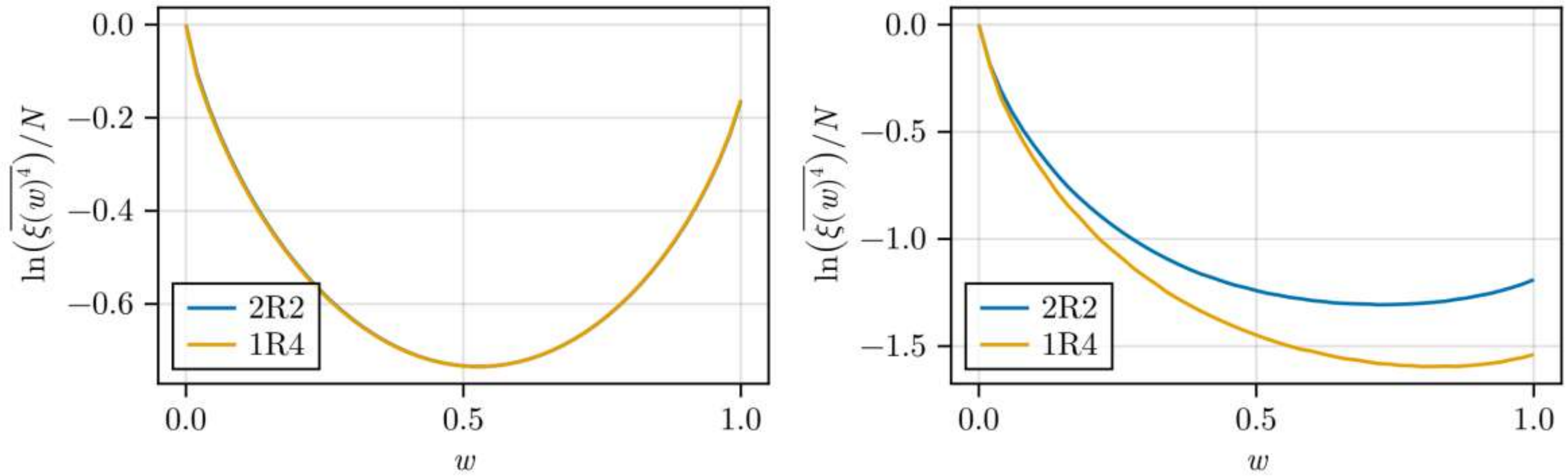
(b) $\beta = 1$



(c) $\beta = 5$



(d) $\beta = 20$

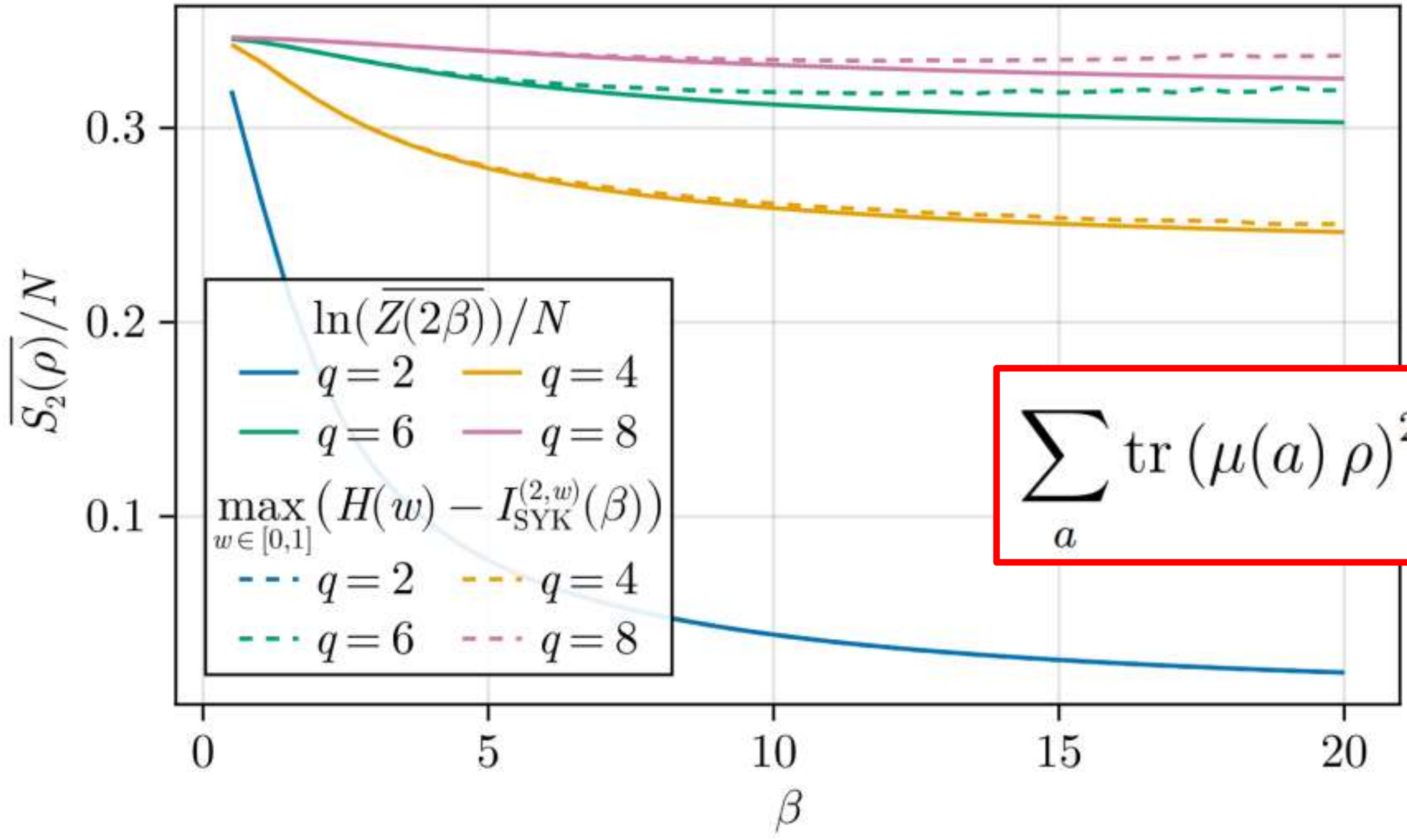


(a) $q = 2$ (b) $q = 4$

enhanced $O(R)$ replica symmetry

R=2 and purity

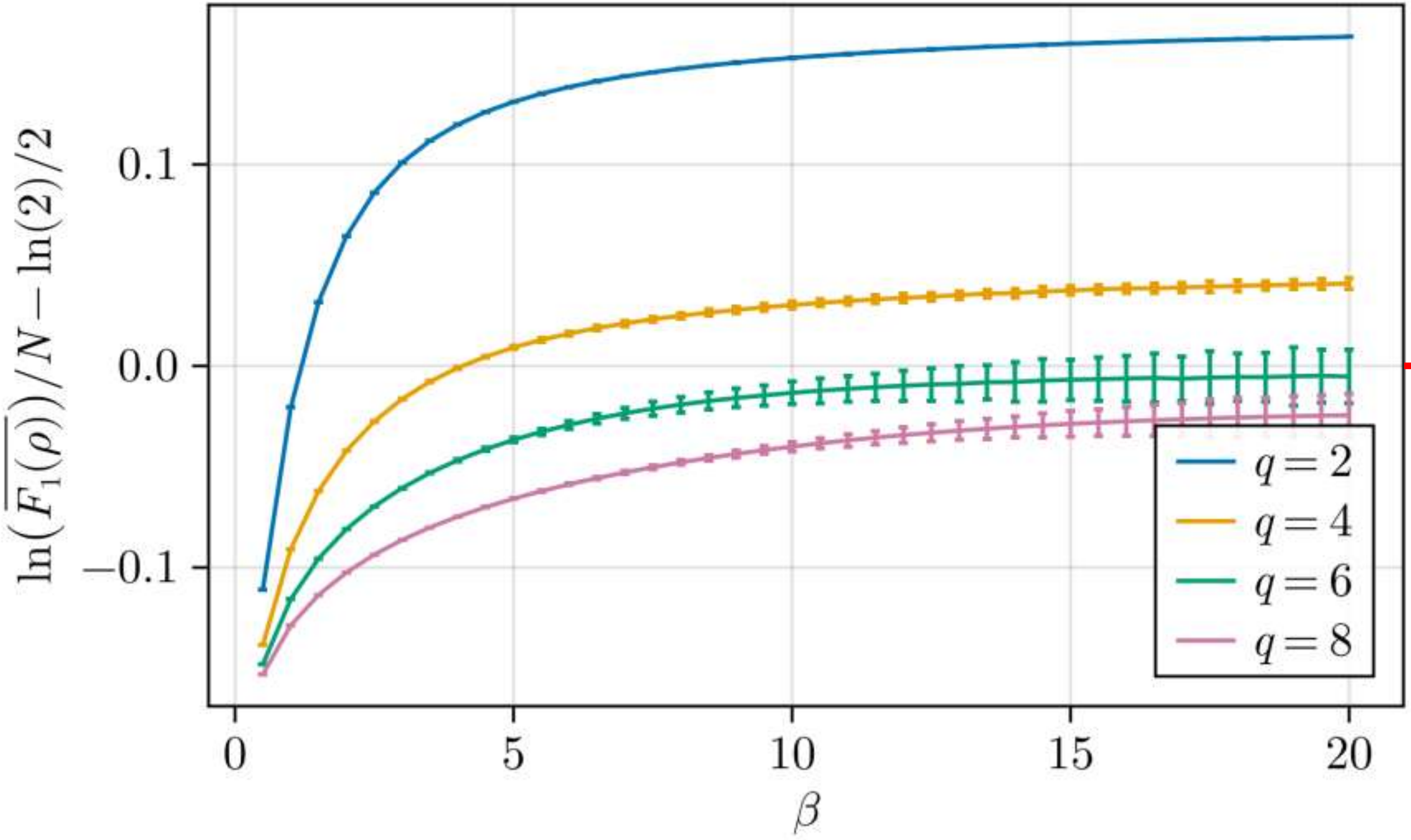
$$\overline{\sum_a \operatorname{tr}\left(\mu(a)\, e^{-\beta H_{\mathrm{SYK}}}\right)^R} \approx \int_0^1 dw \, \exp\left(N\left[H(w) - \mathcal{I}_{\mathrm{SYK}}^{(R,w)}\right]\right)$$



$$\boxed{\sum_a \operatorname{tr}\left(\mu(a)\, \rho\right)^2 = D \operatorname{tr}(\rho^2)}$$

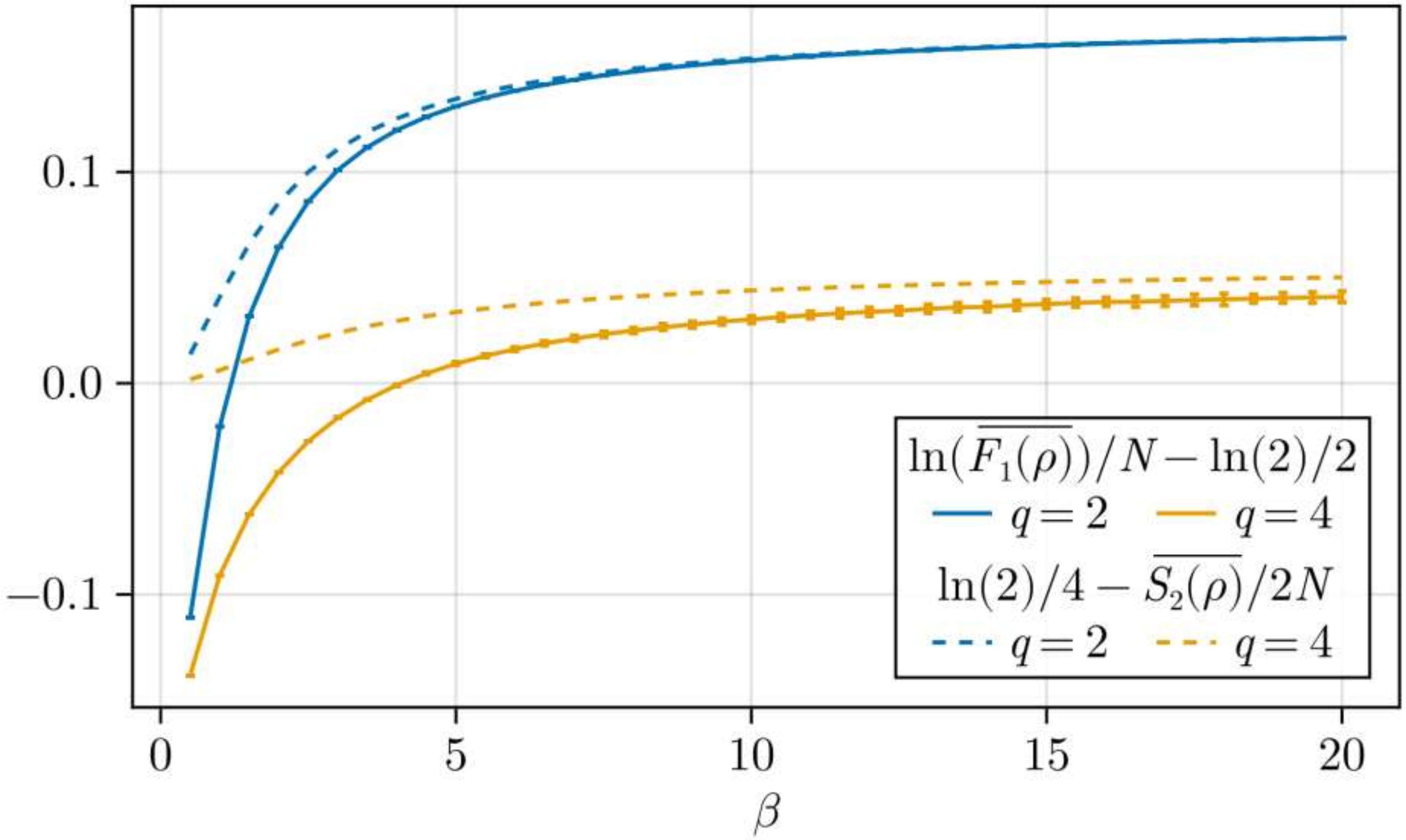
Robustness of magic

$$R(\rho) \geq \frac{F_1(\rho)}{D}, F_1(\rho) = \sum_a |\xi(a)|$$



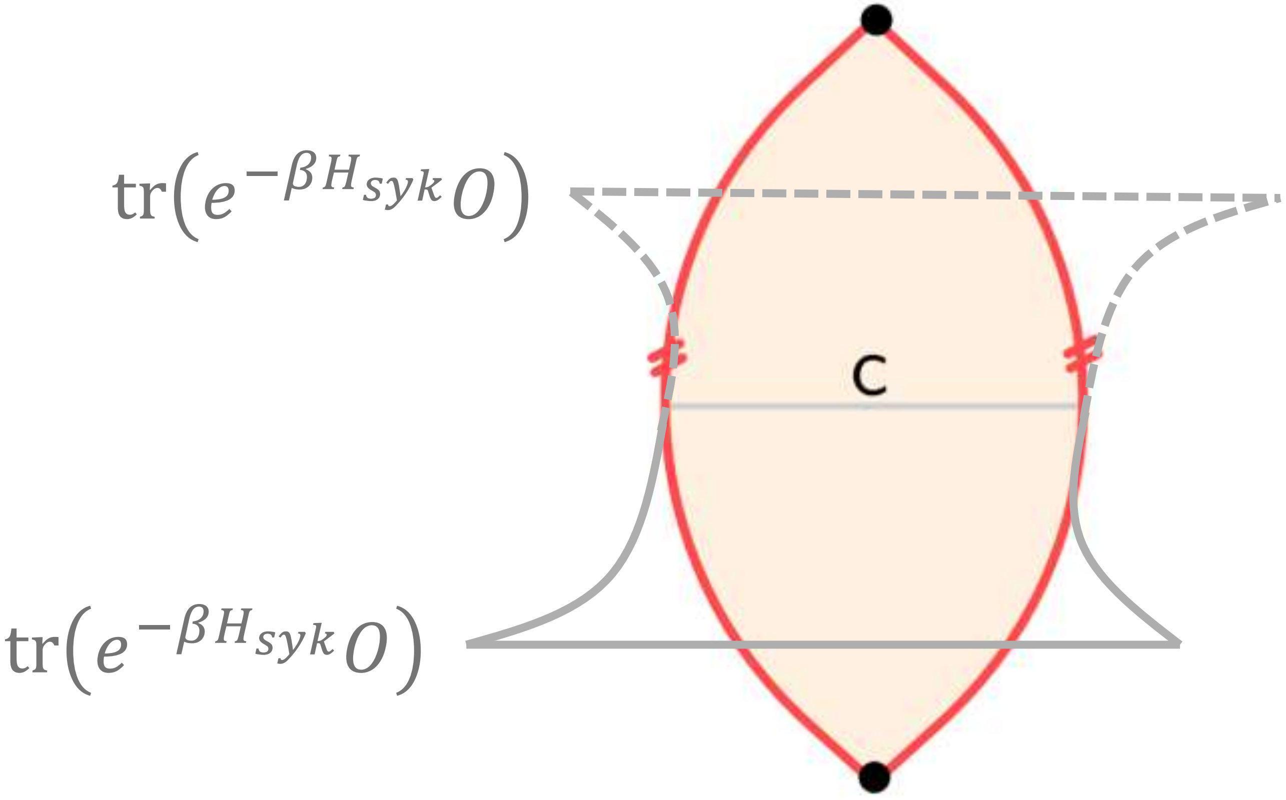
guaranteed highly magical

if strings with w near .5 control the answer, then we showed that the 1-norm obeys: $\frac{\overline{F_1(\rho)}}{D} \approx e^{N[\ln(2)/4 - s_2/2]}$.



Magic and Wormholes

Enter wormholes and closed universes



We model the inter-replica correlations, G_{21}, as correlations between the two sides of a Euclidean wormhole; analytic continuation gives a closed universe [Maldcena-Maoz]

[JT wormhole/cosmology solutions: Stanford, Usatyuk-Wang-Zhao]

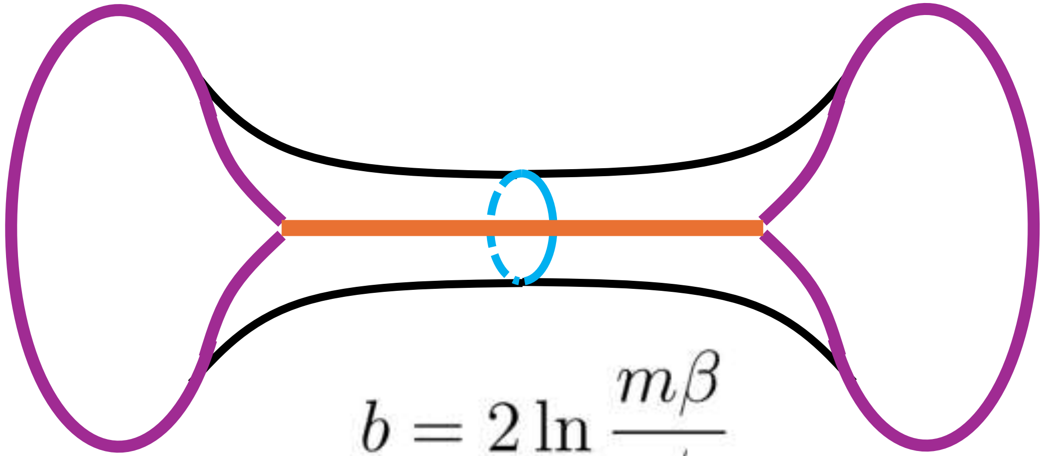
Wormhole calculation

- We model the SYK inter-replica correlation with a boundary-to-boundary correlations of N light probe fermions moving in a background geometry stabilized by a heavy particle [Bettaque-S]
- The bulk 2-point function of the light fermions is computed by summing over geodesics with mass → scaling dimension
- Background – double trumpet

$$ds^2 = d\rho^2 + b^2 \cosh^2 \rho \, d\sigma^2$$

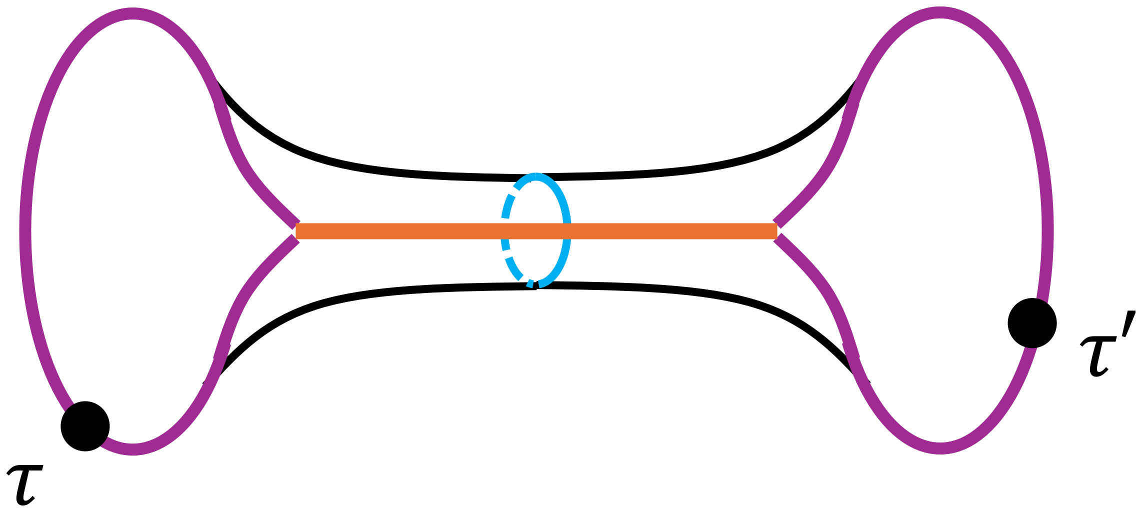
$$\Phi = \frac{\pi \phi_r}{\beta} \cosh \rho \cosh b\sigma.$$

[Stanford, Usatyuk-Wang-Zhao]



$$b = 2 \ln \frac{m\beta}{\pi \phi_r}$$

Wormhole calculation



$$e^{\rho_c} = \frac{2\beta}{\pi\epsilon}\frac{1}{\cosh b\sigma}.$$

$$\frac{\tau(\sigma)}{\beta} = \frac{\arctan e^{b\sigma} - \arctan e^{-b/2}}{\arctan e^{b/2} - \arctan e^{-b/2}}.$$

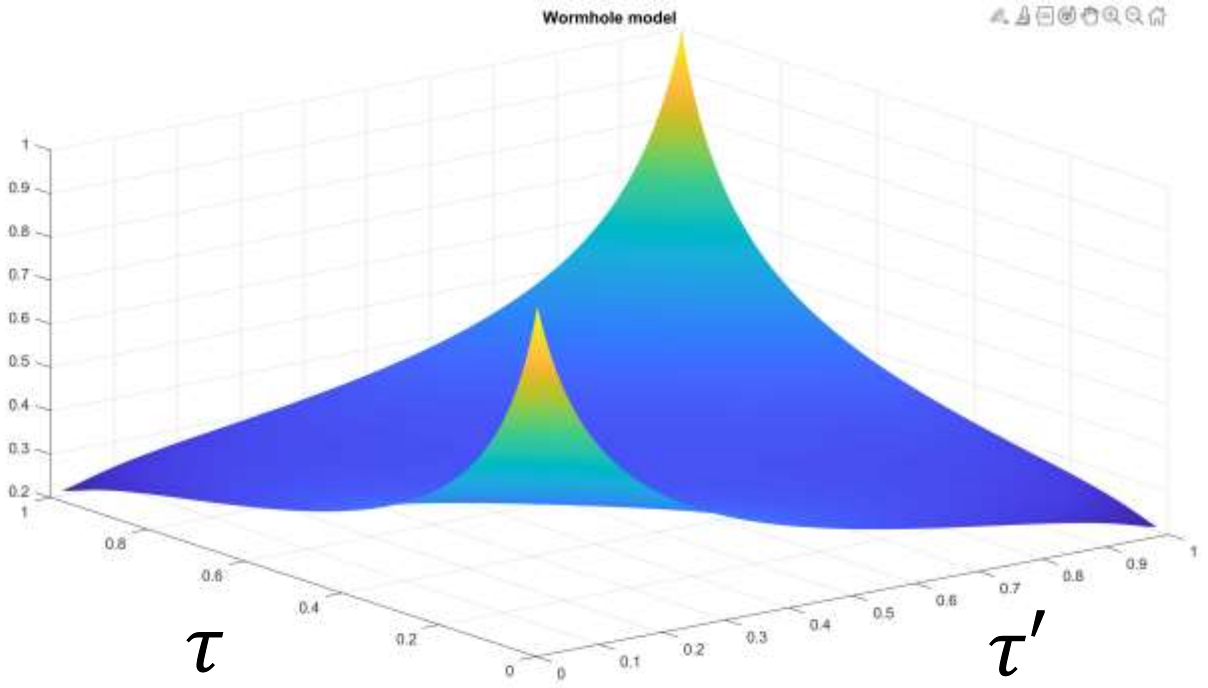
winding geodesics $d_n(\sigma,\sigma') = \rho_c(\sigma) + \rho_c(\sigma') + \log\cosh^2\frac{b(\sigma-\sigma'-n)}{2}.$

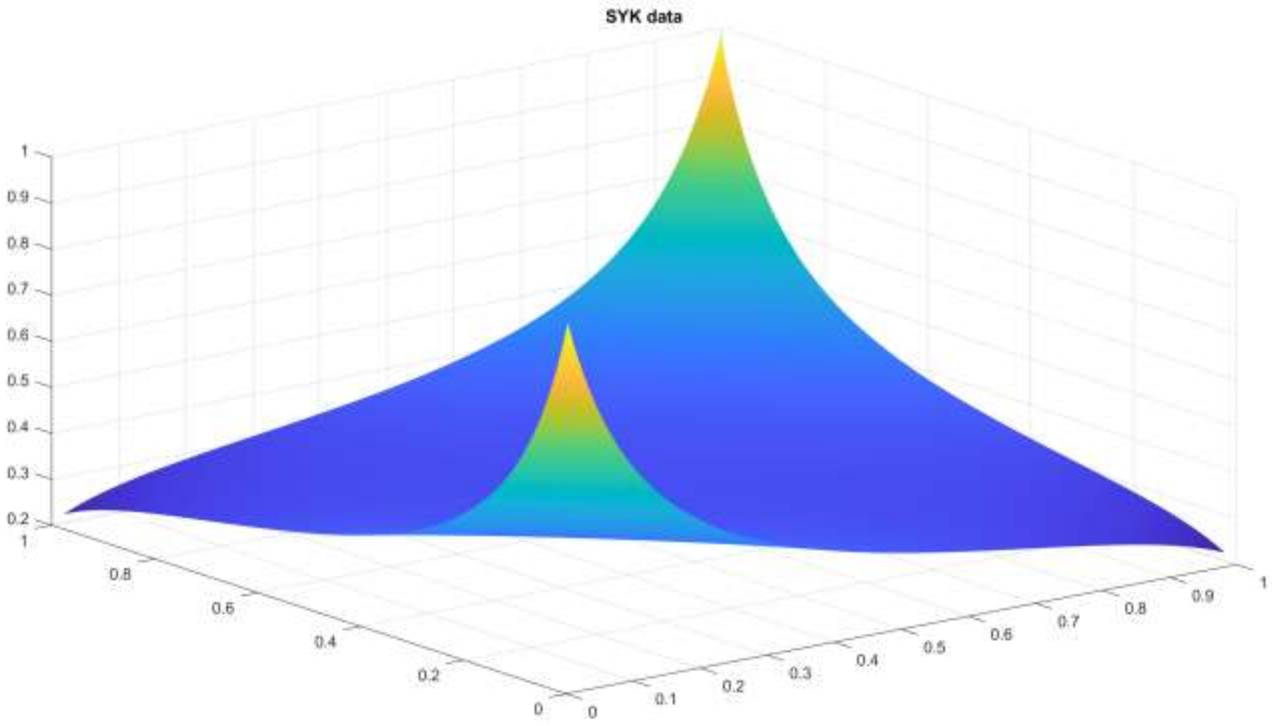
$$\mathcal{G}_P(\tau,\tau') = \mathcal{N}_P\sum_n e^{-\Delta d_n(\sigma(\tau),\sigma'(\tau'))}, \quad \mathcal{G}_{AP}(\tau,\tau') = \mathcal{N}_{AP}\sum_n (-1)^n e^{-\Delta d_n(\sigma(\tau),\sigma'(\tau'))},$$

[Bettaque-S]

Comparison of normalized correlations

$$\mathcal{G}_f = (1-f)\mathcal{G}_{AP} + f\mathcal{G}_P.$$





Bulk model: $b = 6.4, f = .32$; SYK: $w = .25, \beta J = 20, L = 1000$

[Bettaque-S]

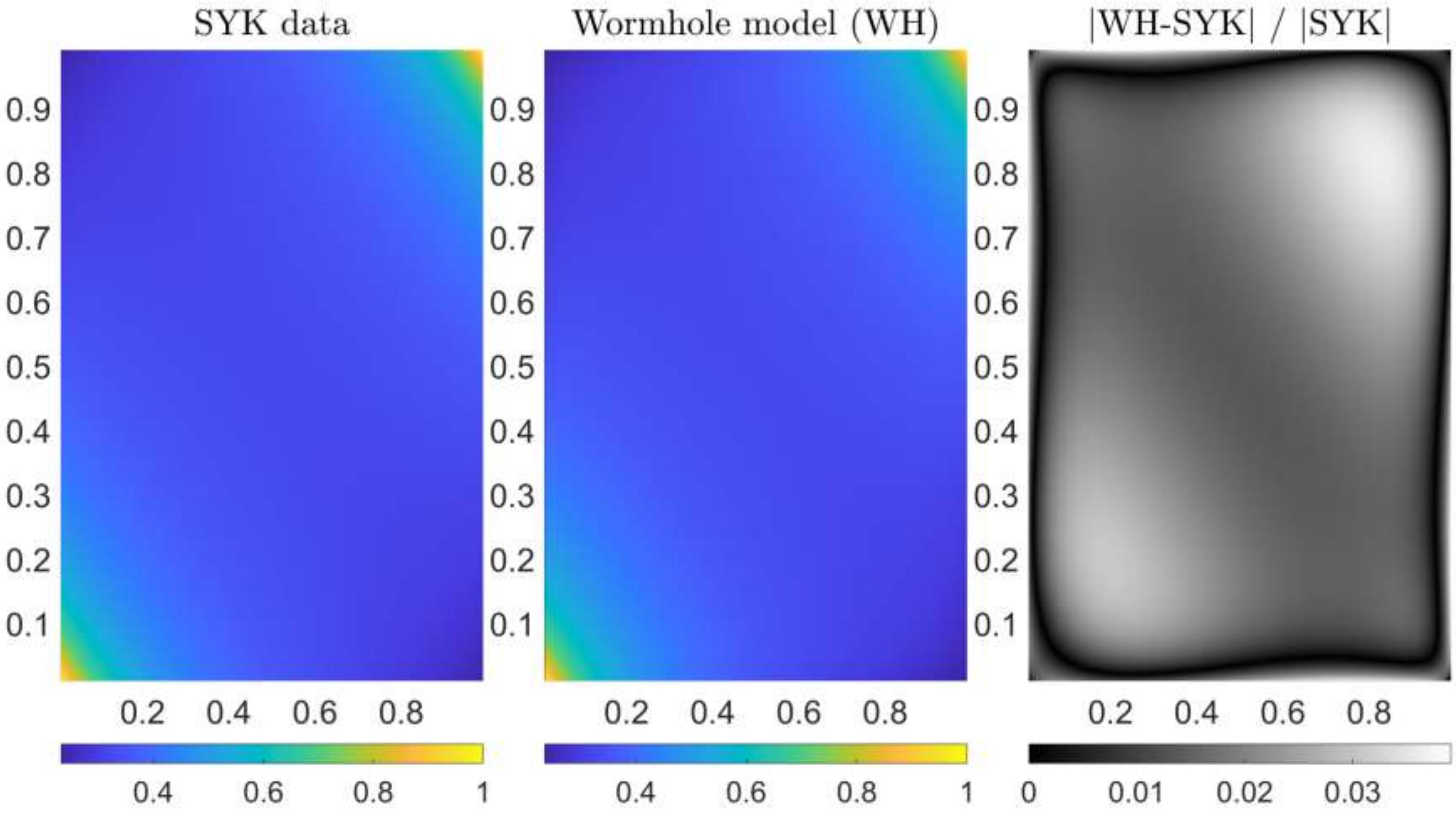
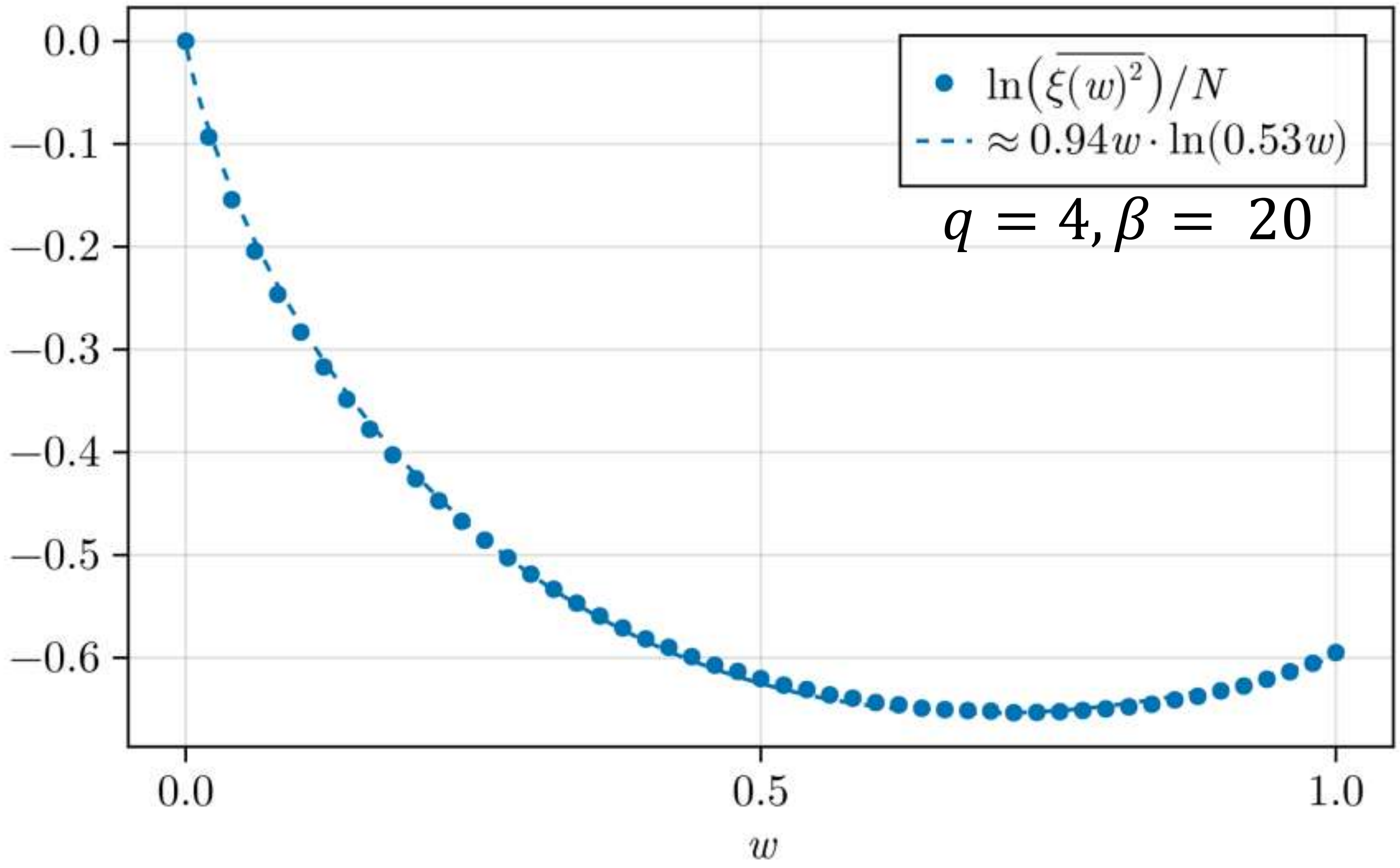


Figure 21: (Left) The plot shows the value of $G_{21}(\tau,\tau')$ in color as a function of τ/β and τ'/β. It is normalized by its maximum value, which occurs in the corners of the square. The SYK parameters are $\beta = 20$, $w = .25$, $L = 2000$. (Middle) The same configuration now showing the normalized wormhole prediction obtained by fitting to the SYK data. The fit values are $b = 6.5$ and $f = .33$. (Right) The absolute value of the relative difference between (Left) and (Middle). The maximum deviation is around 4%.

[Bettaque-S]

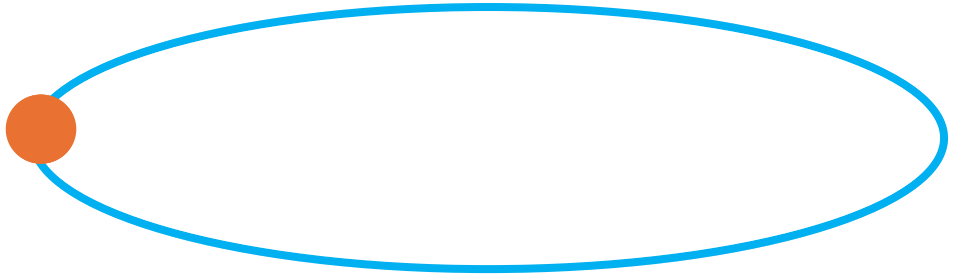
Action data

$$I_{\mathrm{JT}} = -S_0\chi - \frac{1}{2}\int d^2x\sqrt{g}\Phi(R+2) + (\text{boundary terms}) - m\ell_H + I_{\mathrm{matter}}$$

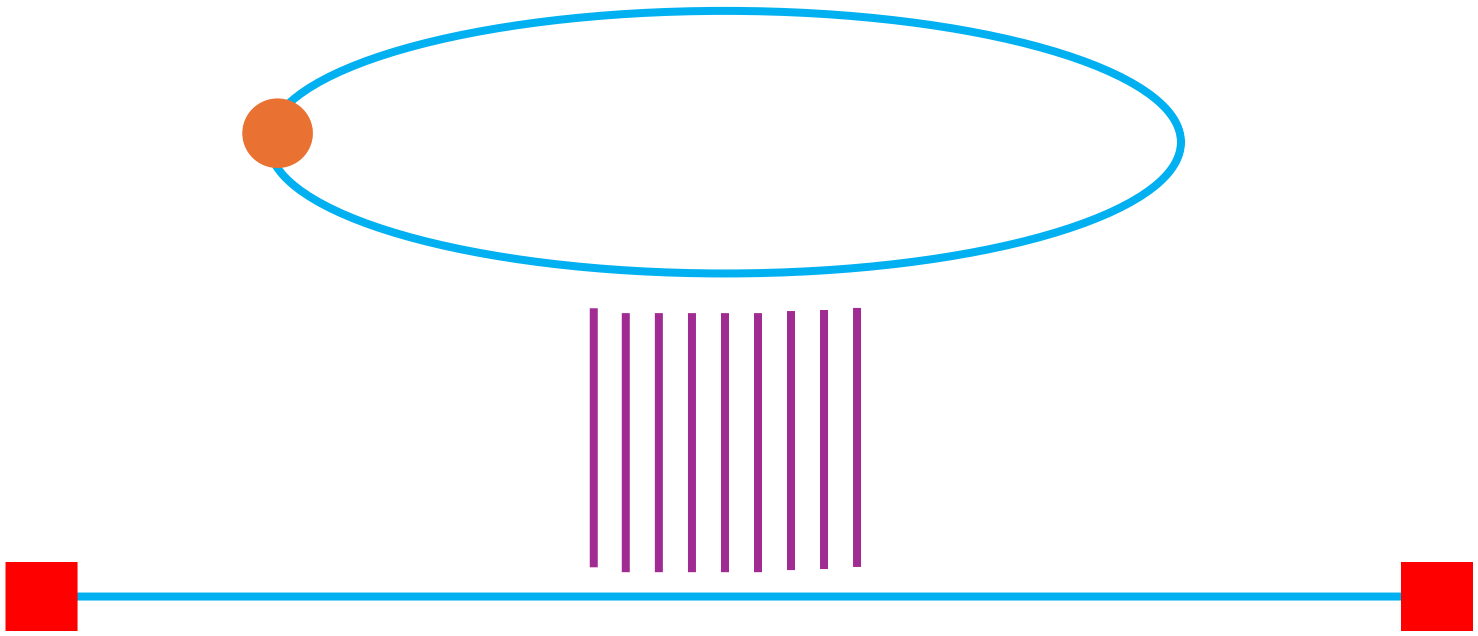


Baby universes from SYK

The baby universe



The space of the closed universe consists of a mostly empty circle with a heavy clump of matter



There are also quantum fields in the closed universe which can be entangled with fields in a separate spacetime region with asymptotic AdS boundaries → this gives a state in the Hilbert space associated with the boundaries

[a version of AS^2: Antonini-Sasieta-S, ...]

Microscopic model

- We start with the Sachdev-Ye-Kitaev (SYK) model on N fermions:

$$\{\psi_i, \psi_j\} = 2\delta_{ij}$$

$$H_{\mathrm{SYK}} = \mathrm{i}^{\frac{q}{2}} \sum_{i_1 < \ldots < i_q} J_{i_1 \ldots i_q} \psi_{i_1} \ldots \psi_{i_q}$$

$$\overline{J_{i_1 \ldots i_q} J_{i'_1 \ldots i'_q}} = \frac{\mathcal{J}^2 N}{2q^2 \binom{N}{q}} \delta_{i_1, i'_1} \ldots \delta_{i_q, i'_q}$$

Microscopic model

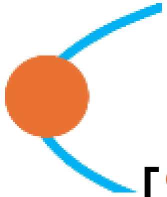
- We also need two additional ingredients:
  - (i) A first order phase transition to effectively evaporate a black hole, thus disconnecting its interior which becomes the closed universe
  - (ii) A heavy operator to support the closed universe

For (i), we use the Maldacena-Qi Hamiltonian:

$$H = H_{\mathrm{SYK,L}} + H^{*}_{\mathrm{SYK,R}} - \frac{\mu N}{\binom{N}{p}} (-\mathrm{i})^{p^2} \sum_{i_1 < \ldots < i_p} \psi^{\mathsf{L}}_{i_1 \ldots i_p} \psi^{\mathsf{R}}_{i_1 \ldots i_p}$$

For (ii), we use a many-fermion operator:

$$\mathcal{O}_{\Delta} = \mathrm{i}^{\frac{r}{2}} \psi_{i_1} \ldots \psi_{i_r}$$



[Sasieta-S-Vilar Lopez]

Microscopic state

heavy operator acting on left side of
maximally entangled LR state

$$|\mathcal{O}_\Delta\rangle \equiv \mathcal{O}^{\mathsf{L}}_\Delta |I\rangle$$

- Thermal pure state:

$$|\Psi_\beta^\Delta\rangle \propto e^{-\frac{\beta}{2}H} |\mathcal{O}_\Delta\rangle = \sum_n e^{-\frac{\beta}{2}E_n} c_n |E_n\rangle$$

thermal cooling with
MQ Hamiltonian

erratic coefficients

energy eigenstates
of MQ Hamiltonian

- This is the microscopic state of the coupled SYK dots; it varies erratically with the SYK couplings and the choice of operator and its statistical properties encode the closed universe physics

[Sasieta-S-Vilar Lopez]

What do we compute?

$$\mathcal{N}_\beta^\Delta = \langle \mathcal{O}_\Delta | \, e^{-\beta H} \, | \mathcal{O}_\Delta \rangle = \sum_n e^{-\beta E_n} |c_n|^2$$

ensemble average $$\overline{\mathcal{N}_\beta^\Delta} \sim e^{-I_\beta[\Sigma_\mathrm{d}]} Z_\mathrm{d} + e^{-I_\beta[\Sigma_\mathrm{h}]} Z_\mathrm{h}$$

We can develop a path integral expression for this quantity either directly in the SYK language or using a corresponding bulk theory

[Sasieta-S-Vilar Lopez]

Semi-classical bulk picture

JT gravity coupled to matter, action: $I = I_{\mathrm{JT}} + I_{\mathrm{matter}}$

$$I_{\mathrm{JT}} = -S_0\,\chi[\Sigma] - \frac{1}{2}\int_{\Sigma} \Phi(R+2) - \int_{\partial\Sigma} \Phi_b(K-1)$$

Field content:

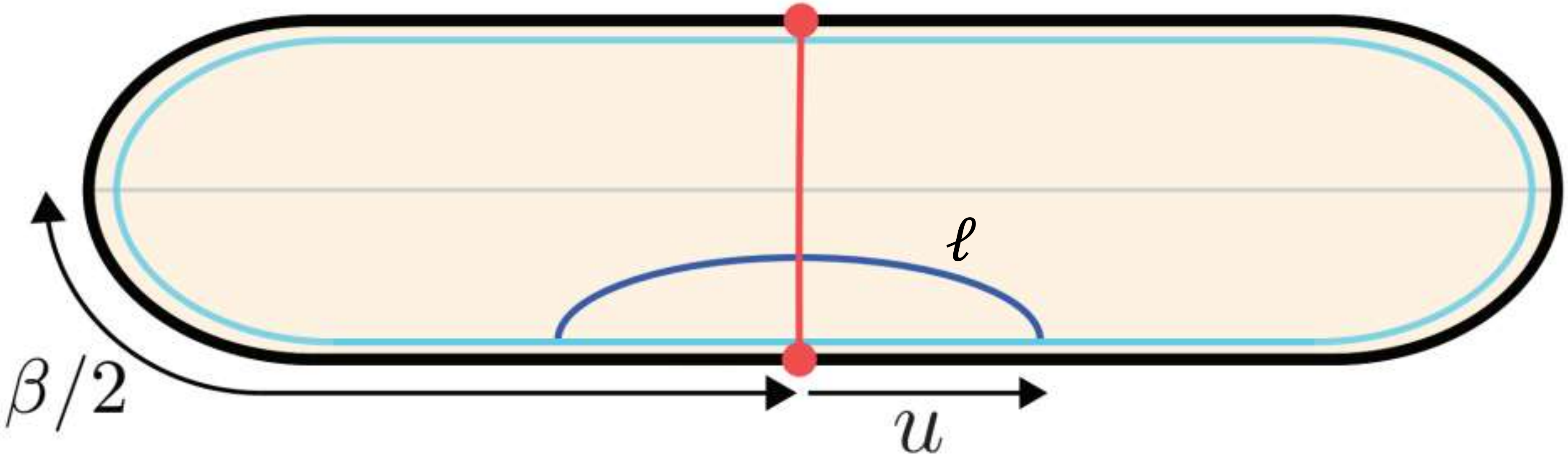
- The metric is fixed to be constant curvature,
- The dilaton (Phi) can have a non-trivial profile
- The matter can be taken to be O(N) massive fermion fields

[Kitaev, Sachdev, Maldacena-Stanford, Rosenhaus-Polchinski,...]

Two saddles

$$\overline{\mathcal{N}_\beta^\Delta} \sim e^{-I_\beta[\Sigma_\mathrm{d}]} Z_\mathrm{d} + e^{-I_\beta[\Sigma_\mathrm{h}]} Z_\mathrm{h}$$

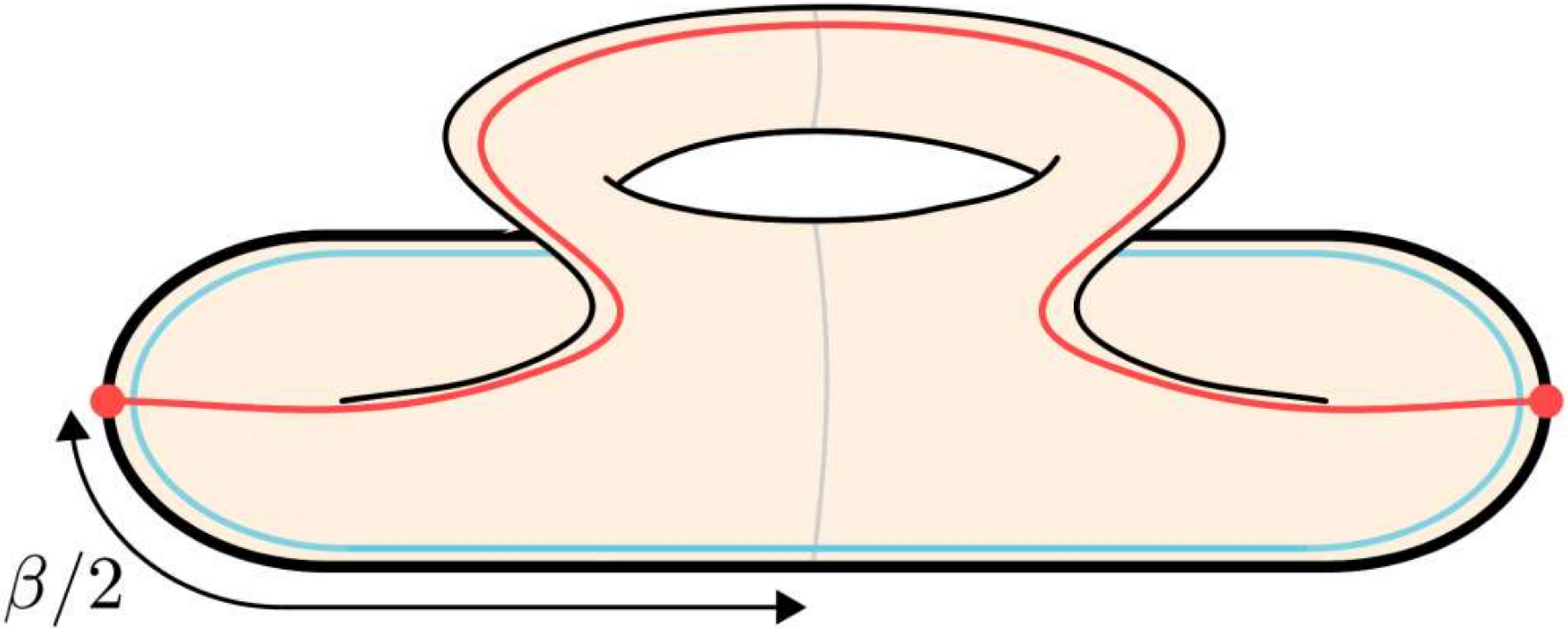
- Black hole saddle (Euclidean)



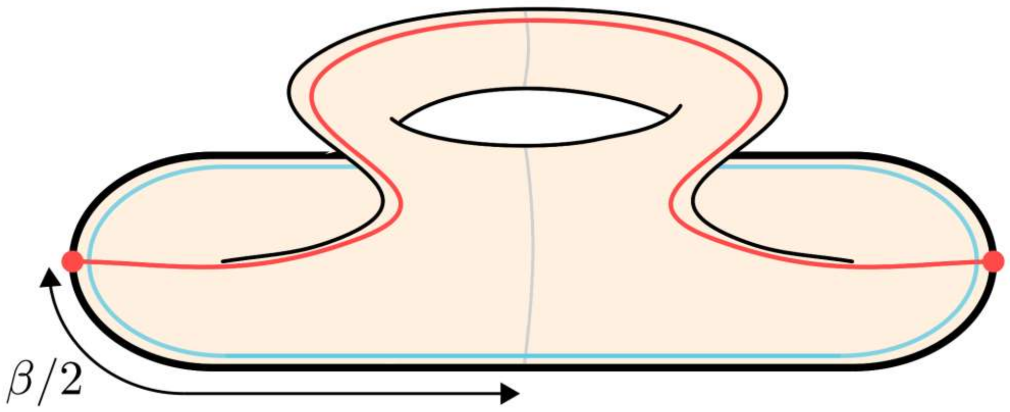
$$I = \frac{N}{2}\int \mathrm{d}u \left(\frac{1}{2}\dot{\ell}^2 + V(\ell)\right) - S_0 \qquad V(\ell) = 2e^{-\ell} - \eta e^{-\Delta_H \ell} + \frac{2\Delta}{N} e^{-\ell/2}$$

[Maldacena-Qi, Sasieta-S-Vilar Lopez]

- Baby universe saddle (Euclidean)

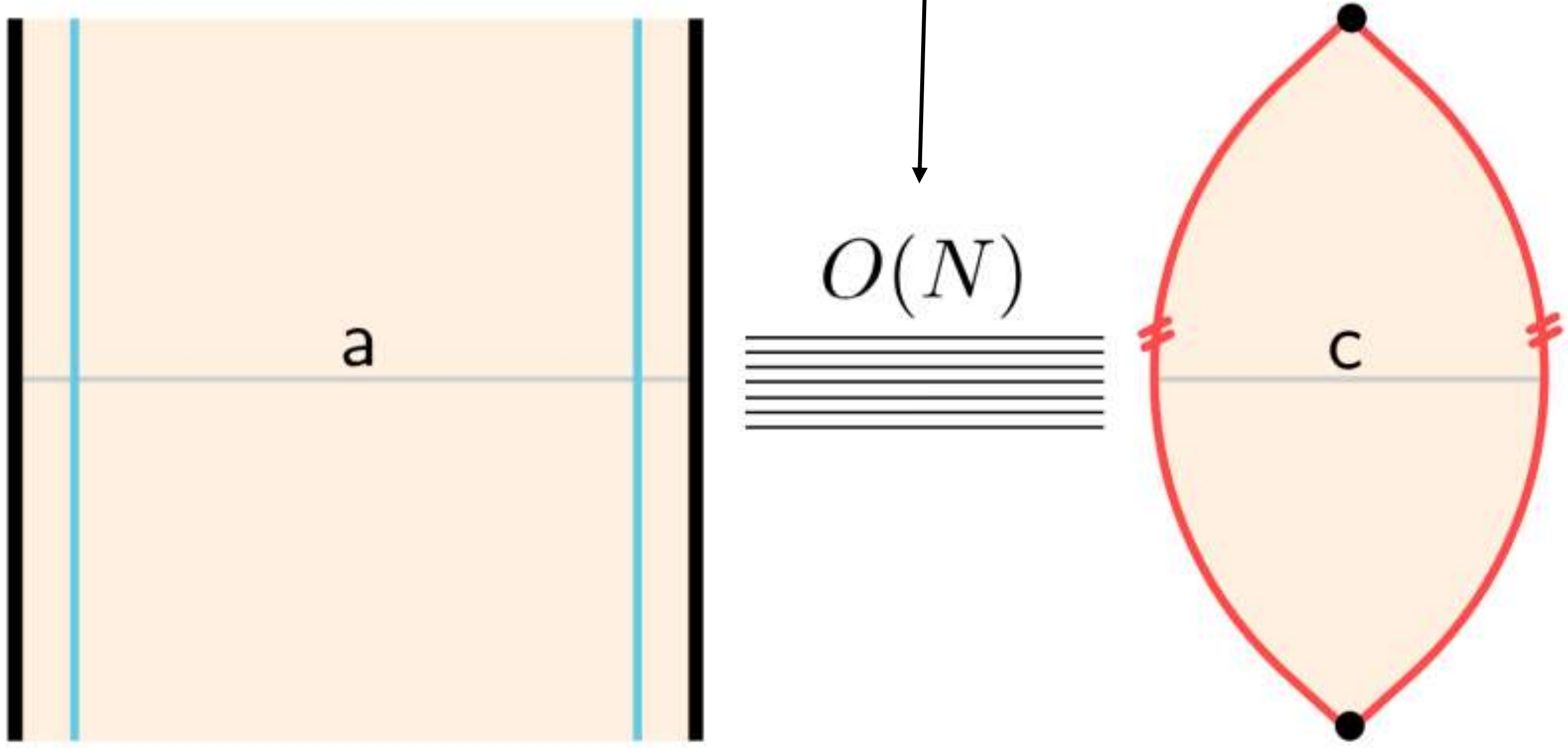


[Sasieta-S-Vilar Lopez]



one way to interpret the entanglement with the closed universe is in terms of the entropy of the ensemble-averaged state

Lorentzian picture:



Zero MQ temperature: an even simpler limit?

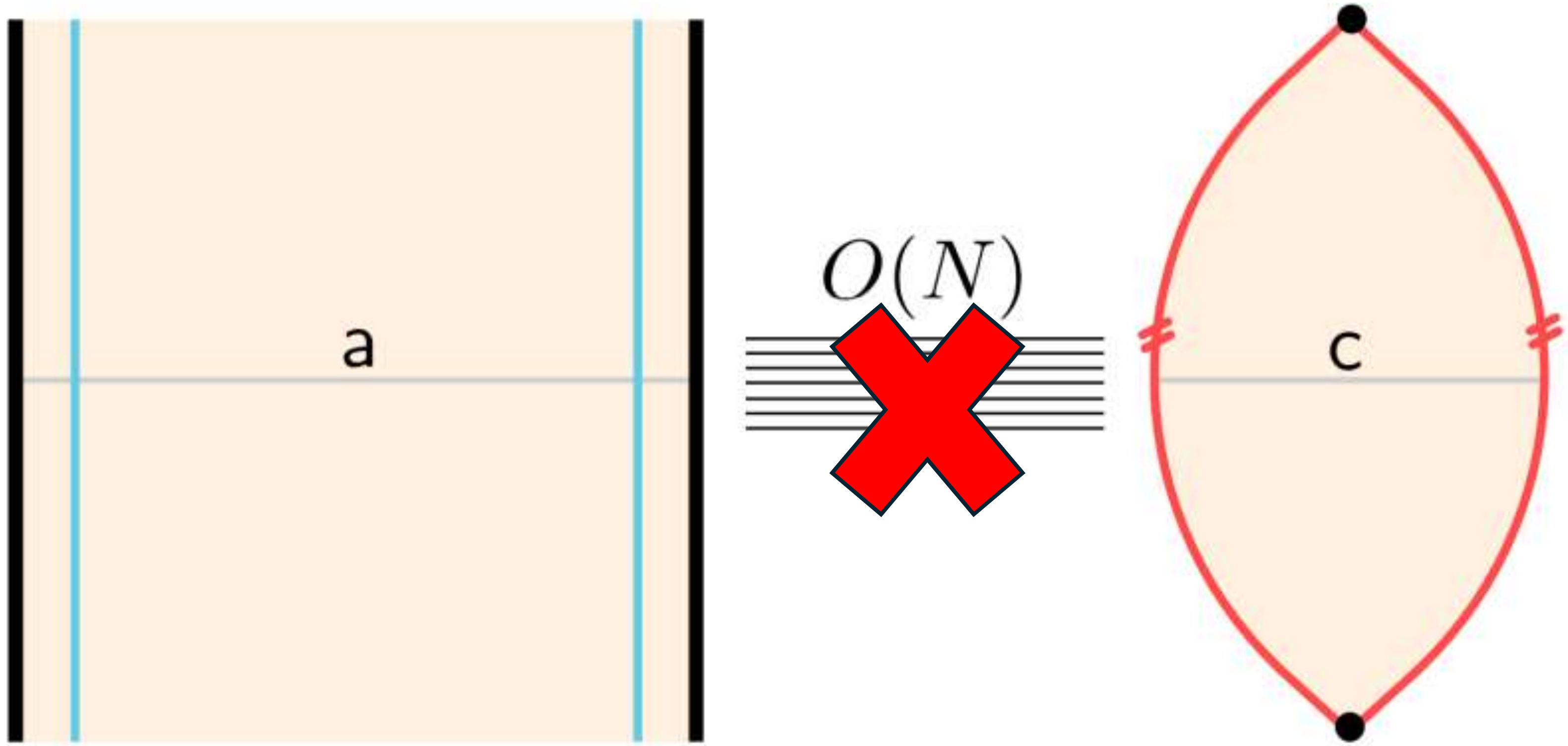
- If we take the beta parameter to infinity, then the microscopic state becomes proportional to the MQ ground state
- The MQ ground state is close to the thermofield double (TFD) state of the original SYK at a coupling dependent temperature

$$H = H_{\mathrm{SYK,L}} + H^*_{\mathrm{SYK,R}} - \frac{\mu N}{\binom{N}{p}} (-\mathrm{i})^{p^2} \sum_{i_1 < \ldots < i_p} \psi^{\mathrm{L}}_{i_1 \ldots i_p} \psi^{\mathrm{R}}_{i_1 \ldots i_p}$$

[Maldacena-Qi, Cottrell et al.]

- The SYK TFD is not erratic and isn't dual to a closed universe???

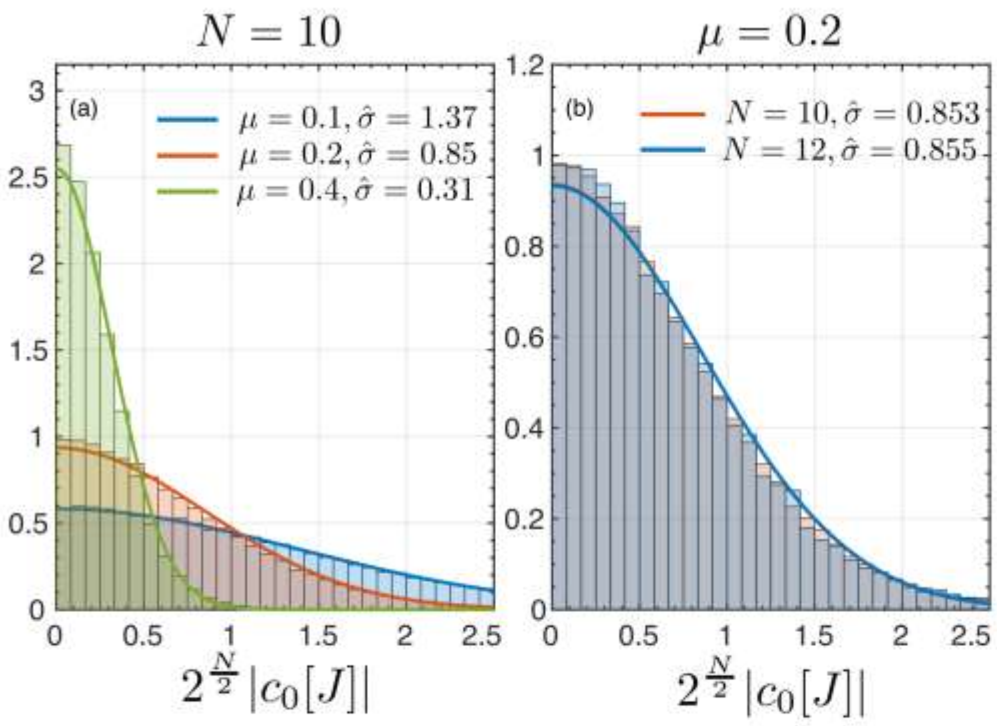
What's going on the bulk?



We're zeroing out the bulk entanglement with the closed universe, so the closed universe is floating "untethered" in the void

There is still something erratic …

- The overlap between the MQ ground state and the heavy operator state is erratic → the closed universe is hiding in the statistical properties of this matrix element, which is a thermal 1-pt function!



Small N data shows that the overlap behaves as a real gaussian random variable, we saw the same at large N

[Sasieta-S-Vilar Lopez, Bettaque-S]

Summary and outlook

- We can compute measures of magic in the SYK model using path integrals, giving one of the very few windows into magic at large N
- We found that the SYK data at low temperature could be quantitively fit with a gravity computation involving wormholes in JT + matter, giving a kind of holographic dual of magic
- Our results also connect to holographic models of closed universes, with the SYK results providing a first demonstration of the heavy operator randomness hypothesis of AS^2
- There is a great deal to do!

Comments on cosmology

It from Bit???

- We have a match between averaged quantities in a microscopic theory and cosmological solutions in semi-classical gravity
- The erratic nature of the microscopic matrix elements and statistical averaging was crucial to reveal the closed universe
- This is a very active field at the moment, e.g. observers, debates about saddles, semiclassicality, …

[averaging/algebras: Liu, Kudler Flam-Witten, Gesteau-Engelhardt, Gesteau, observers: Engelhardt-Gesteau-Harlow, Abdalla-Antonini-Iliesiu-Levine, Harlow-Usatyuk-Zhao, saddles: Vilar Lopez-Van Raamsdonk, averaging/semiclassicality: Antonini-…-S, large q SYK …]